

SE-ACBiG: An Explainable CNN-BiGRU Framework with Channel Attention for Plant Disease Detection

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CANDIDATE'S DECLARATION

We Karthi Abhilash Kumar (2K22/SE/86) and Muaviz Maroof Khan (2K22/SE/116) students of B.Tech Software Engineering, hereby declare that the Project titled “SE-ACBiG: An Explainable CNN-BiGRU Framework with Channel Attention for Plant Disease Detection and Sustainable Agricultural Monitoring” submitted by us to the Department of Software Engineering, DTU, Delhi, in partial fulfillment of the requirements for the award of the Bachelor of Technology degree, is an original work carried out by us under proper guidance and supervision. This work has not been copied from any source without appropriate citation and acknowledgment. Furthermore, this work has not previously formed the basis for the award of any Degree, Diploma, Fellowship, or any other similar title or recognition.

Place: New Delhi
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CERTIFICATE

I hereby certify that Project titled “SE-ACBiG: An Explainable CNN-BiGRU Framework with Channel Attention for Plant Disease Detection and Sustainable Agricultural Monitoring” which submitted by Karthi Abhilash Kumar (2K22/SE/86) and Muaviz Maroof Khan (2K22/SE/116) for fulfillment of the requirement for awarding of the degree of Bachelor of Technology (B.tech) is a record of the project work carried out by the students under my guidance & supervision. To the best of my knowledge, this work has not been submitted in any part or fulfillment for any Degree or Diploma to this University or elsewhere.

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ABSTRACT

The thesis proposes an innovative explainable deep learning approach, called SE-ACBiG, for intelligent and sustainable plant disease detection and diagnosis. The method uses SqueezeExcitation (SE) channel attention block, Convolutional. Neural Network (CNN), Bidirectional Gated Recurrent Units (BiGRU), multi-head attention mechanism, and Grad-CAM visualization technique to enhance both prediction accuracy and model interpretability. In contrast to traditional deep learning methods, SE-ACBiG offers a visualization of predictions through Grad-CAM maps which enables localization of the infected leaf areas.

To overcome important gaps and limitations in existing literature, the thesis uses a systematic ten-step preprocessing approach which involves image validation, normalization, class distribution analysis, imbalance assessment, oversampling, train-test split, and data augmentation. The dataset imbalance is explicitly identified and fixed in the thesis through oversampling with an imbalance ratio of 3.07. Moreover, five different baselines, namely, Basic CNN, CNN+LSTM, CNN+BiGRU, CNN+BiGRU+Attention, and ACBiG-mLiW, have been used to compare the performance of the proposed model framework, SE-ACBiG.

One of the major contribution to the presented research lies the focus on XAI and methodology correctness. TAmong the examined SE-based frameworks, SE-ACBiG is the only one that includes feature recalibration in SE, the localization by Grad-CAM for disease identification, balanced training, correction of statistical evaluation metrics, AUC-ROC analysis, and cross-metric rank evaluation. The proposed solution is more explainable, reproducible and applicable to sustainable agriculture, and outperform other framework in terms of disease detection. Prospective research will be directed towards the expansion of this suggested framework to include edge IoT devices, UAV crop inspection, and the detection of diseases on major crops, like wheat and rice.

We also propose using the Grad-CAM module as a visual explainability tool in our architecture to mark the affected areas of the leaf as a result of diseases.

The examples of the proposed SE-ACBiG model are compared with the other baseline models in the experiments performed on the balanced Plant Village dataset. The suggested method gives an accuracy rate of 99.8%, 98.84% Precision, 98.81% Recall, 98.82% F1-Score, 99.91% AUC-ROC, also has excellent validation accuracy and high class separability, the suggested model produces the heat map using the Grad-CAM method, which enables identifying the affected regions.

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Without acknowledging the contribution of those who have played an important role in making the success possible, the achievement of any kind of accomplishment will be both incomplete and meaningless since their contribution was necessary for its success.

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CHAPTER 1

INTRODUCTION

1.1 Background and the Motivation

Agriculture serves as the economic pillar of developing countries and is crucial for world food security. Even though there have been many technological developments in farming, plant disease still leads to huge losses; according to a report from the Food and Agriculture Organisation, plant pests and diseases are a major responsible for 20-40% of production losses globally each year [9]. Among the diseases that affect tomato, corn, wheat, rice, and apple plants, some of the economically most critical ones can be listed.

Until recently, identifying diseases in plants relied heavily on manual detection through farmers' and domain experts' observations. This process, in addition to being time-consuming and costly, is subjective and not feasible on a larger scale, especially considering the lack of agricultural knowledge in rural areas of developing countries. Fortunately, the development of AI and DL technologies allows for automatic disease detection from leaf images.

The innovative work of Hughes & Salathé [5] was the formation of the PlantVillage database which consists of more than 54,000 leaf images labeled as having one of the 38 diseases on 14 plant species [5]. After creating the database, Mohanty et al. [10] proved that Neural Network can provide a high level of accuracy of 99% under laboratory conditions [10]. The more advanced approaching involved transfer learning for improved efficiency rate using convolutional neural networks like VGG16, ResNet, InceptionV3, DenseNet, MobileNet, and EfficientNet [7, 8, 11].

Rapid development of research in this area has revealed a lot of problems related to the existing techniques, as these techniques only used black box models, and did not consider data imbalance, used poor evaluation methods, and require too much computational power. The main problem with the existing techniques is the fact that they have no way to explain the results meaning a model can provide a results, but cannot show why it did that, which means that farmers have no way to prove or disprove them.

1.2 Limitations of Existing Systems

1. Lack of Explainability of Predictions: Most machine learning algorithms make predictions that are not accompanied by any form of reasoning or visual cues. Therefore, it is impossible for farmers to check whether the algorithms were influenced by disease lesions or irrelevant textures [4].
2. Imbalance between Classes: In many agricultural datasets, especially the PlantVillage dataset, there is an imbalance of classes. Studies often ignore the class balance problem, causing an inherent bias towards dominant classes and overestimation of accuracy [6].
3. Lack of Comprehensive Evaluation Protocols: Accuracy and F1-scores dominate the evaluation process. Other useful performance indicators such as AUC-ROC, RMSE, MAE, MAPE, R^2 score, among others, do not find any mention [6].
4. High Computational Complexity: Transfer learning models based on deep learning networks, such as VGG16 network with more than accounted 138 million parameters, are not computationally feasible for edge devices in agriculture [8].
5. Failure to Localize Disease: Existing classifiers can classify diseases but lack the capability to localize the exact part of the leaf that is infected by diseases [4].
6. No Integration of Attention Mechanism and Sequence Learning: Previous research has used CNN, LSTM, and BiGRU models independently. The proposed model is the first one that incorporates SE channel attention, BiGRU sequence learning, multi-head attention mechanism, and grad-CAM explainability [6].

1.3 Role of Machine Learning and Deep Learning in Agriculture

ML and DL technologies are revolutionizing the agricultural sector through automation and intelligence [13, 14]. Convolutional Neural Networks (CNNs) are useful in learning hierarchical spatial information from images detecting abnormalities in colour, texture, and morphology associated with plant diseases [15]. Sequence models such as LSTM and BiGRU enrich the representations provided by CNNs by learning dependencies within the feature sequences [2]. Attention mechanisms, that includes channel attention using Squeeze-and-Excitation (SE) blocks [1] and global attention using Transformer-style Multi-Head Attention [3], further improve the discriminative advantage of deep learning models by allowing for the weighting of feature channels and sequence positions based on their importance.

Explainable Artificial Intelligence (XAI), notably Grad-CAM [4], helps close the performance-gap that is inherent to artificial intelligence models through visual explanations understandable to the subject matter expert. The applications of AI to agriculture encompass but are not limited to crop yield prediction, soil quality assessment, smart irrigation, pest detection, weather predictions, precision fertilization, unmanned aerial vehicles field monitoring, and agricultural robotics [13, 14, 16]. See Figure 1.1 for a conceptual illustration of these AI applications.

Figure 1.1: Application of Artificial Intelligence and Deep Learning in Smart Agriculture

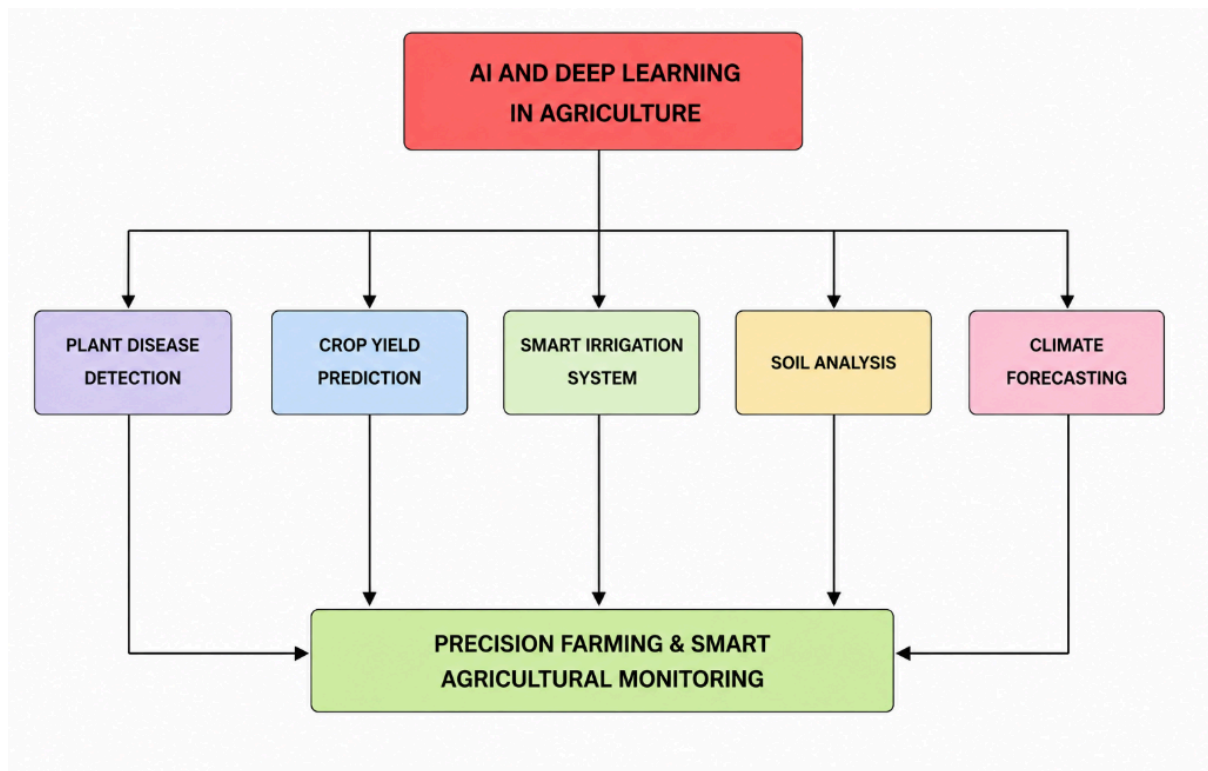


Figure 1.1: Application Areas for AI and Deep Learning in Current Agricultural Practices. The figure shows that AI technologies like disease identification, yield prediction, precise irrigation, soil analysis, and unmanned aerial vehicle (UAV) surveys all play crucial roles in enabling smart and sustainable agricultural systems.

1.4 The Proposed SE-ACBiG Framework

It is an SE-ACBiG architecture designed for addressing all the above mentioned drawbacks and comprises of:

- Feature extraction based on plant leaf images using a hierarchical spatial method based on CNN and
- Application of Squeeze- Excitation (SE) channel attention method [1] to dynamically re-calibrate features, remove background noise and enhance disease related channels
- The use of the BiGRU [2] model that enabled learning of forward and backwards dependencies of the feature sequence
- Feature global interactions through. Multi-Head Attention [3] that does allows attention on disease specific regions
- Preprocessing pipeline which consists of ten stages include imbalance detection, oversampling, augmentation, and stratification; and

- Evaluation of metrics such as AUC-ROC, RMSE, MAE, MAPE, R^2 , validation loss, confusion matrices.

1.5 Research Objectives

1. The main goal of the proposed project is develop an efficient, and computationally effective architecture of a deep learning model for the intelligent recognition of plant diseases. This research is aimed at achieving the following:
2. For multiclass classification of plant diseases, CNN layer, SE channel attention, BiGRU layer, and Multi-Head Attention technique are used to design a SE-ACBiG neural networks structure.
3. To design a tenstep image preprocessing pipeline including image validation, normalization, quantification of class imbalances, oversampling, encoding, stratified splitting, and data augmentation
4. To perform a Grad-CAM explain in order to visualise the results and explain the logic of AI decision-making [4].
5. The remaining 5 baselines -Basic CNN, CNN-LSTM, CNN-BiGRU, CNN-BiGRU + Attention and ACBiG-mLiW [6]-are compared to the SE-ACBiG algorithm.
6. To evaluate the computational costt and the inference time for deployement AI models for farming.
7. However, when further investigating agricultural deep learning, it is necessary to pay attention to problems that occurred in previous research [6] and also to determine the method of proper evaluation.

1.6 Scope of the Study

The SE-ACBiG method will allow for the use of experiments conducted on five different disease types that are used in the PlantVillage (Early Blight of Tomato, Late Disease of Tomato, Healthy Tomato, Black Rot of Apple, Common Rust of Corns) while maintaining a balanced and realistic data set.

Some of the scope limitation include: conducting experiments with only five PlantVillage classes, using images captured under conditions, and no testing of practical implementation of the methodology in the field with actual physical IoT devices.

1.7 Thesis Organisation

Structure of this thesis will be as the followed: The Chapter 2 discusses on current literatures related to plant disease detection, deep learning techniques, attention-based models, and XAI amongst others. The novel SE-ACBiG method is introduced in Chapter 3, including details of the preprocessing, architecture, training and Grad-CAM explainability. Chapter 4 lists the experiment, along with the setup of the system, the software/hardware configuration, the setup of the datasets and the procedure of the experiments. Experimental results and comparisons are presented in chapter 5. In Chapter 6, the architectural lessons learned from the experiment, methodological contributions, limitations, and deployment aspects are discussed. The thesis is followed by a conclusion on which the conclusion in Chapter 7 hinges.

CHAPTER 2

LITERATURE REVIEW

2.1 Deep Learning for Plant Disease Detection

Since with the emergence of deep learning approaches to detecting diseases on plants, there has been tremendous development due to the availability of big labelled data sets and the improvement in GPU computing. One example of such a data set is the PlantVillage data set, which was published by Hughes and Salathé (2015) [5]. This is considered as the basic data set in the domain, and includes more than accounted 54,000 labeled leaf images from 38 disease classes and 14 different plants. Figure 2.1 show some image from this dataset.

Fig. 2.1 Sample images from PlantDoc and Soil Types datasets.

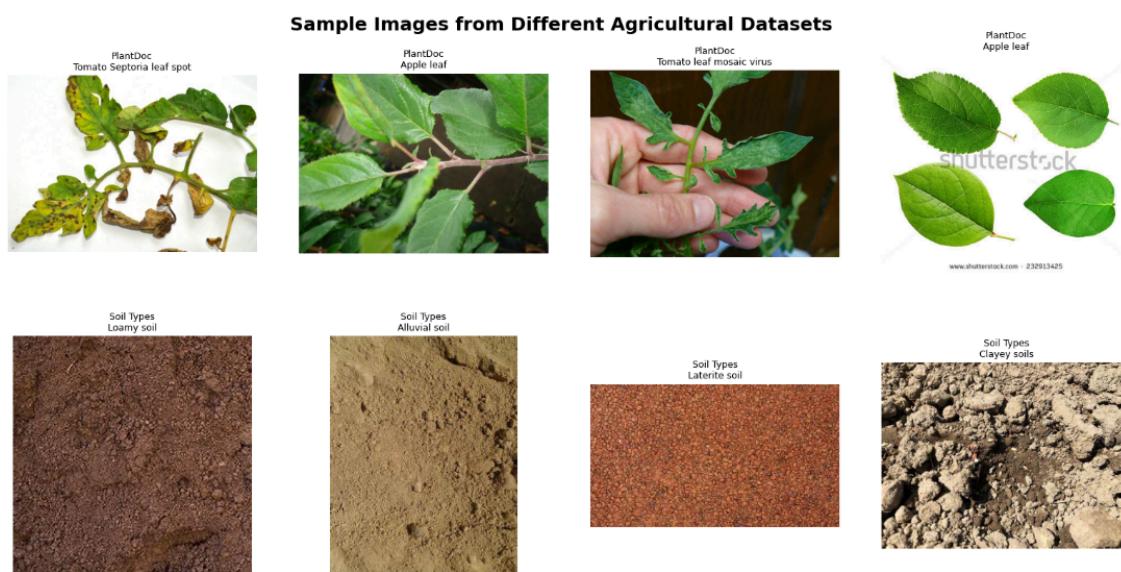
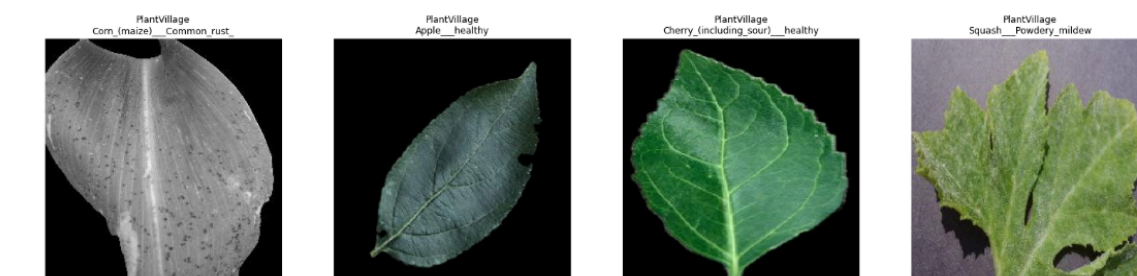


Fig. 2.2. Sample healthy and diseased leaf images from PlantVillage dataset.



Sample images from Plantillage Dataset [5] are shown in Fig 2.1 and 2.2. This.. figure represents health and diseases affected leaves regions images for Tomato (Early Blight and Late Blight), Apple (Black Rot) and Corn (Common Rust) revealing the visaul diversity and class similarty challenging the classification system.

Application of transfer learning in plant disease detection model through CNNs showed rapid. CNN model such as VGG16, ResNet-50, InceptionV3, DenseNet-121, MobileNetV2 [7] and EfficientNet-B0 [8] were found to be more efficient and converged well when using a limited dataset. All of them, apart from MobileNetV2 and EfficientNet-B0 models, are used to reduce model parameters and pay attention to their application in edge computing. Sarma et al 2022 [13] explained the utilization of CNN-based disease detection model integrated IoT monitoring devices in the development of the advanced precision agriculture. systems. Nagaraj et al. 2024 [15] provided a comprehensive survey of the pest detection models which are based on CNN.

Low level of explainability as black box model, always need class balanced data set and neglect of multi-metric evaluations frameworks like AUC-ROC curve and regression based metrics are among common disadvantages of all the transfer learning models applicable to the proposed model. Despite high accuracy, all transfer learning models are subjected to common disadvantages, which apply to the proposed method too: they lack explainability as black box models, always assume class-balanced datasets and do not consider multi-metric evaluations frameworks encompassing AUC-ROC and regression-based metrics.

2.2 Sequential and Hybrid Deep Learning Architectures

Identification of the presence of locally structured plant diseases and contextual dependencies led to the design of architectures that combine CNNs with recurrent elements [17]. LSTM-based hybrid approaches became some of the pioneering attempts to highlight the advantages of modelling features sequentially over the simple CNNs.

The work of Mendoza-Bernal et al. (2024) [12] presents an anomaly detection system using CNN with data augmentation techniques in smart agriculture application. Despite high levels of classification accuracy, this method does not include aspects of sequential modelling, imbalance, and explainability. A method by Wongchai et al. (2022) [17] used attention-based deep models with RNNs for intelligent farming. Temporal features could be effectively learned but the high computational complexity of the approach due to the use of simple RNNs and no disease localization capability.

Introduction of GRUs as a less computationally expensive version of LSTMs [2], needing lesser parameters to have a comparable level of representational capacity, paved the way for research into the Bidirectional GRU architecture. BiGRU takes care of the contextual dependencies, which unidirectional LSTM is unable to handle. The research paper by Sellam et al. (2025) [6] has introduced ACBiG-mLiW that uses deep CNN layers with BiGRU, Multi-head attention and Modified Leaf in Wind (mLiW) optimizer, attaining an accuracy in the range of 97.5%-98.5% in various agricultural environments.

However, ACBiG-mLiW suffers from many conceptual flaws as per this review paper: lack of explainability method; using only 144 soil images in one experiment out of four, leading to invalid statistics; using statistical test in only one experiment out of four; a completely wrong formula for specificity; lack of AUC-ROC evaluation; no handling of class imbalance problem; and lack of confusion matrix and cross-metric ranking. Such a flawed methodology severely questions the validity of results reported by it, and forms the methodological basis for this paper.

2.3 Attention Mechanism in Deep Learning

Attention mechanism did gain significance in advanced deep learning frameworks by helping models focus selectively on the important parts of the input data. Vaswani et al. (2017) [3] developed the concept of the Transformer network along with Multi-Head Attention mechanism to show that attention alone, sans recurrence, can surpass baselines on sequence modeling tasks. Multi-Head Attention connects queries, keys, values into the several subspaces, thereby allowing for simultaneous attention to several representational aspects of the input.

With respect to tasking of plant disease detection, the use of attention mechanisms helps serve two functions: improvement in the performance of classification through exclusion of background information and provision of an inherent level of interpretation due to attention on particular regions. By incorporating the concept of Multi-Head Attention in CNN-BiGRU architecture, we are able to combine local information (CNN) with contextual sequential information (BiGRU) and attention-based global information, something which is impossible with just one component baselines.

2.4 Squeeze-and-Excitation Networks

Hu et al. (2018, 2020) [1] did introduce the concept of SE Networks as a lightweight channel attention module designed to model inter-channel dependencies in order to enable adaptive recalibration of feature maps. The SE block works in three sequential steps:

Squeeze: The Global Average Pooling operation which decrease spatial feature maps to channel descriptors of size C , representing global channel information.

Excitation: Two connected layers equipped with ReLU and Sigmoid activation functions generate channel importance weights from the descriptors, thus modelling channel-level non-linear interaction.

Scale: Element-wise multiplication of learned weights with the input feature maps helps recalibrating channel contributions - de-emphasizing irrelevant channels such as background textures, and focusing on disease-related ones.

SE blocks offer great gains in performance despite having a very low computational cost in applications such as ImageNet classification, medical image analysis, object detection, and fine-grained image categorization [1]. Although effective for image classification tasks in general, channel attention using SE blocks had never before been combined with CNN-BiGRU hybrid architecture in agriculture disease identification tasks.

2.5 Explainable AI in Agricultural Disease Detection

Grad-CAM [4], introduced by Selvaraju et al., calculates the weighted sum of the gradients backpropagated from the predicted class scores to the final convolutions to produce visual attention heatmaps, which mark regions with high positive gradient weights in warm colours (i.e., red, yellow), indicating the regions found discriminative by the network.

Grad-CAM has been successfully applied in the diagnosis of diseases in medical images [18], for tumour localisation and defect detection in industrial settings, and in autonomous systems. Although effective, there have been few applications in agriculture disease detection systems. The reason is that the majority of published works are black-box AI models trained primarily for accuracy, thus leaving agricultural experts unable to evaluate the effectiveness of these models due to lack of

explanation capability. This issue is the key reason why current disease diagnosis tools are not readily adopted in practice.

Recent papers have already shown that explainable recommendations are highly demanded, e.g. in fertilisation tasks [16] and crop management [19]. The current paper addresses this issue by adopting Grad-CAM in the design of SE-ACBiG.

2.6 Research Gaps

The systematic literature review reveals the following critical gaps that motivate the present research:

1. There is currently no plant disease detection system that can offer all of the following characteristics: classification accuracy, explainability, balance, SE channel attention, sequential bidirectionality, metric-based performance evaluation, and efficient computational performance within one framework.
2. The best-performing hybrid approach, ACBiG-mLiW [6], suffers from flaws in methodology (improper calculation of specificity, lack of AUC-ROC, no imbalance correction, lack of confusion matrices) which render its results unreliable.
3. No research work has combined Squeeze-and-Excitation mechanism channel attention with CNN-BiGRU models for agriculture-related disease detection.
4. Gradient-weighted Class Activation Mapping (Grad-CAM) used by researchers within their plant disease detection framework as a means of model explainability rather than a requirement.
5. Cross-metric ranking is a rigorous approach to holistic model comparisons that is yet to be applied to agricultural disease detection deep learning algorithms.

Figure 2.3 summarises the identified gaps and illustrates how SE-ACBiG addresses each limitation.

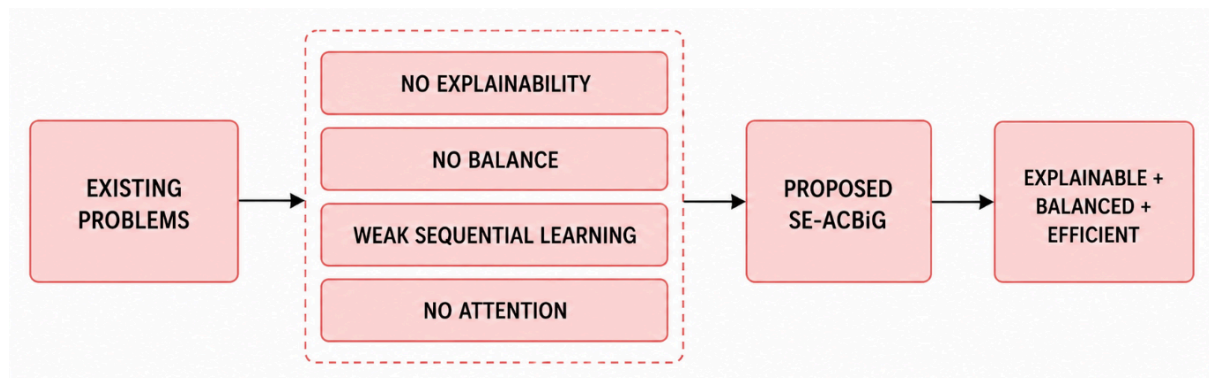


Figure 2.3: Research Gap Analysis and Motivation for the SE-ACBiG Framework. The figure maps identified limitations in existing plant disease detection systems including absent explainability, class imbalance, insufficient evaluation, and lack of integrated attention to the corresponding SE-ACBiG architectural and methodological contributions.

2.7 Literature Review Summary

The chapter provided an overview of plant disease detection which is based on deep learning techniques, hybrid CNN and recurrent architectures, attention mechanisms, Squeeze-and-Excitation network, and explainable artificial intelligence techniques. Important insights from this literature

review are that: (1) CNN-based transfer learning is prevalent in the area but fails to offer explanations and comprehensive evaluations; (2) BiGRU performs better than LSTM in sequential contexts and is computationally less expensive; (3) SE channel attention works well in terms of recalibrating features at low computational costs and has never been used in agriculture-related CNN-BiGRU architectures; (4) Grad-CAM is the most appropriate XAI technique for localization purposes yet has been scarcely included in disease detection models; and (5) there is a lack of an integrated framework that combines the above five aspects together.

CHAPTER 03

PROPOSED METHODOLOGY

3.1 Overview

The chapter discusses the full methods and approach for the SE-ACBiG model, including data set description and selection, the ten-stage pre-processing pipeline, baseline architectural considerations, the newly designed SE-ACBiG architecture, training setup, and explainability using Grad-CAM.

The architectural structure of the framework is shown in Figure 3.1 below.

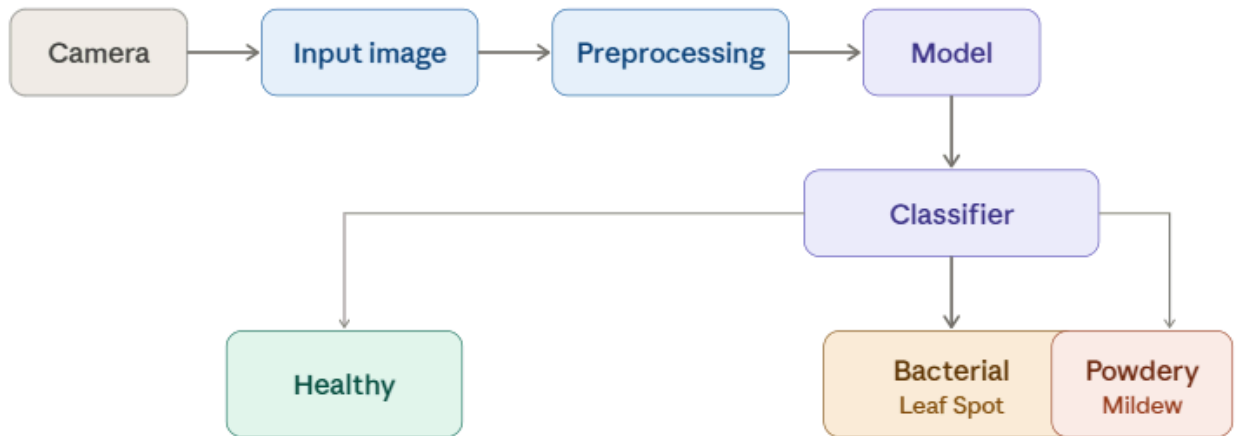


Figure 3.1: System Architecture of the Propose SE-ACBiG Framework. This figure portrays the whole system architecture from the input of the leaf images to the output of the Grad-CAM heatmap for the identification of disease location.

3.2 Dataset Description and Class Selection

The experiment were performed with the Plantillage dataset [5] which comprises 54306 RGB images of leaves, annotated with 38 diseases, over 14 plants. From observed these disease, chose five relevant diseases considering their weight in agriculture and feasibility regarding class imbalance as follows:

Table 3.1: Selected Disease Classes and Initial Sample Distribution from the PlantVillage Dataset [5]

Disease Class	Initial Samples	Test Samples
Tomato Early Blight	800	200
Tomato Late Blight	1,527	382
Tomato Healthy	1,273	318
Apple Black Rot	497	124
Corn Common Rust	953	239
Total / Imbalance Ratio	5,050 / 3.07	1,263

Table 3.1 shows that there is an imbalance ratio of 3.07 (Tomato Late Blight class majority, 1,527 images; Apple Black Rot class minority, 497 images). There is a need for balancing the data set before training the machine learning model. This is demonstrated in Figure 3.2.

3.3 Ten-Step Preprocessing Pipeline

A ten-step preprocessing pipeline is implemented to ensure data quality, address imbalance, and maximise model generalisation. Algorithm 3.1 formally specifies this pipeline.

Algorithm 3.1: Ten-Step Preprocessing Pipeline

Algorithm 1 Plant Disease Dataset Preprocessing and Data Balancing Process

Input: Original raw image folders

$D = \{d_1, d_2, d_3, d_4, d_5\}$ corresponding to five diseases

Output: Balanced and normalized dataset

$(X_{\text{train}}, y_{\text{train}}), (X_{\text{test}}, y_{\text{test}})$

- 1: Select five disease classes,
- 2: Load the RGB images (128×128)
- 3: Remove any invalid images
- 4: Normalize all images: $X \leftarrow X / 255.0$
- 5: Apply one hot encoding to the labels
- 6: Split the dataset into:
 - Train = 80%
 - Test = 20%
- 7: Applied following augmentation: rotation, flip, zoom
- 8: Oversample any of minority classes
- 9: Create a sure balanced dataset
- 10: Return the training and testing sets

. Step 7 measures imbalance explicitly before Step 10 makes corrections to it, something that is not found in ACBiG-mLiW's baseline [6]. This augmentation technique in Step 10 takes into account the real-life scenarios of images taken from agricultural environments.

3.4 Baseline Architectures

Five baselines are built to facilitate systematic evaluation of performances and demonstrate the incrementality of architectures:

Model 1: Simple CNN architecture having three convolutional blocks (with filters: 32, 64, 128 respectively) containing Conv2D → BN → ReLU → MaxPooling → Dropout(0.25). Then by a Global Average Pooling layer before Dense(128) → Softmax(5). This model serves as a benchmark of spatial features extractors.

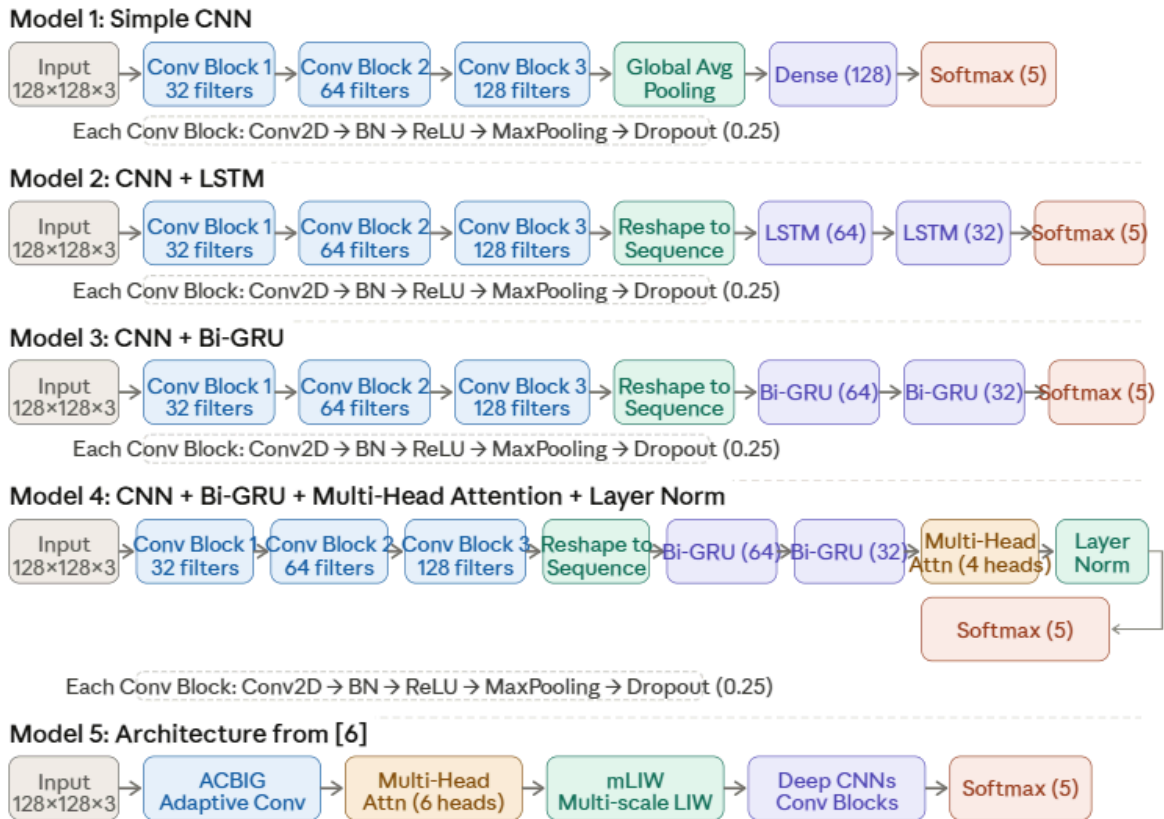
Model 2: CNN+LSTM architecture replacing the classification layer of the above model by two stacked LSTM layers (units: 64, 32) and hence performs sequential analysis of feature sequences.

Model 3: Same as Model 2, except the LSTM layers replaced by bi-directional GRU layers (units: 64, 32) to process both sides of feature sequences simultaneously, something not possible with unidirectional LSTM [2].

Model 4: An improvement on Model 3, consisting of Multi-Head Attention layer with heads=4 and Layer Normalisation, which allows all features to interact globally, something that is not possible with previous models.

Model 5: Architecture reproduced from [6], consisting of ACBiG, 8-headed Multi-Head Attention, mLiW, and deep CNNs.

Figure 4.1: Comparative Architecture of Baseline Deep Learning Models for Plant Disease Detection



The above Figure 4.1 shows the architectural comparison of five deep learning models employed in the classification of plant diseases. As depicted, the architectural evolution starts from a simple CNN model to a more sophisticated sequence-based and attention-based models such as LSTM, Bi-GRU,

multi-head attention, and ACBiG models. It clearly depicts the shift in the process from feature extraction to sequence and interaction.

3.5 Proposed SE-ACBiG Architecture

In SE-ACBiG architecture, SE channel attention blocks are placed between each CNN feature extraction layer and the next layer, thus improving the base architecture of CNN+BiGRU+Attention.

3.5.1 CNN Feature Extraction Block

Four convolutional layers having filters with increasing filter depths of 32, 64, 128, and 256 extract a spatial feature hierarchy from the input leaf image. The four layers have the following structure: Conv2D (3×3, padding=same) → Batch Normalization → ReLU Activation → MaxPooling (2×2) → Dropout (0.25). Thus, lower-level features for textures are captured in early layers, while higher-level disease-specific features are learned in deeper layers.

3.5.2 Squeeze-and-Excitation Attention Block

SE Block [1] is deployed after every CNN block, following the three-step method described in Algorithm 3.2. The reduction factor of $r = 16$ is used in the fully connected layers (FC_1: C/r neurons; FC_2: C neurons). The main contribution compared to Model 4 is that SE Blocks can learn more discriminative features, thus better calibrating features and generating high-quality Grad-CAM heat maps.

3.5.3 Bidirectional GRU Layers

The output from the last SE-CNN layer is then flattened to form a sequence tensor and fed through two layers of BiGRU [2], with 128 and 64 neurons each. Both the BiGRU layers operate on the sequence in the forward as well as backward direction, and the outputs are concatenated. This proposed model outperforms the baseline model using CNN + LSTM in terms of F1 score by 15.9%.

3.5.4 Multi-Head Attention

Output from the BiGRU is first fed to an 8-head Multi-Head Attention layer [3] with a key size of 64 per head, and then follows the Layer Normalisation and residual connections. This module allows the model to be focused on multiple positions within the input sequences in parallel, thereby allowing learning of relationships between disease patterns globally.

3.5.5 Classifier

Dense Layers with three consecutive numbers of neurons (256-128-64) followed by three Dropout Layers each having dropout rates of 0.3, 0.2, and 0.1 respectively help in dimensional reduction until the very last layer which is the Softmax Layer with five neurons

3.6 Training Configuration

All six models (five baselines + SE-ACBiG) are trained using the following standardised configuration:

Table 3.2: Standardised Training Configuration

Hyperparameter	Value / Setting
Optimiser	Adam (Kingma & Ba, 2014) [20]
Initial Learning Rate	0.001
Loss Function	Categorical Cross-Entropy
Batch Size	32
Maximum Epochs	100 (with early stopping)
Early Stopping	Monitor: val_loss; Patience: 15 epochs; Restore best weights
Learning Rate Schedule	ReduceLROnPlateau: factor=0.3, patience=5, min_lr=1e-6
Input Image Resolution	128 × 128 × 3 (RGB)
Random Seed	Fixed for reproducibility across all experiments

Table 3.2 of the training settings and the values used for different hyperparameters in all experimental models. Choice of these settings makes possible stable learning, faster convergence, and less overfitting while training. Same training conditions allow making sure that SE-ACBiG outperforms other methods in a fair manner.

3.7 Grad-CAM Disease Localisation

Algorithm 3.2 (Lines 14-18) presents the Grad-CAM [4] algorithm. Specifically, the algorithm is carried out as: (1) Given input image X and output class c , the forward pass in SE-ACBiG gives the class score $\logit y_c$ and feature maps from the last convolutional layer $F_{\{256\}}$; (2) gradients $\partial y_c / \partial F_{\{256\}}^{\{k\}}$ are obtained via backpropagation for each channel k ; (3) global average pooling of the gradients along spatial axes (i, j) provides the channel-specific weightings α_k ; (4) weighted summation of feature maps with ReLU activation yields the raw heatmap H ; (5) heatmap H is bilinearly upsampled to input size (128×128) and plotted against input image in the form of JET colormap, where red-to-yellow indicates the high discriminative response of the disease. Regions colored by warm colors correspond to observable disease infections (such as fungal infection spots, bacterial blight spots, rust infection pustules) in the input image, thus visually supporting the classification results for farmers and agricultural researchers.

3.8 Framework Workflow

Figure 3.2 contains a comprehensive flowchart of the SE-ACBiG framework pipeline for disease prediction based on imaging data.

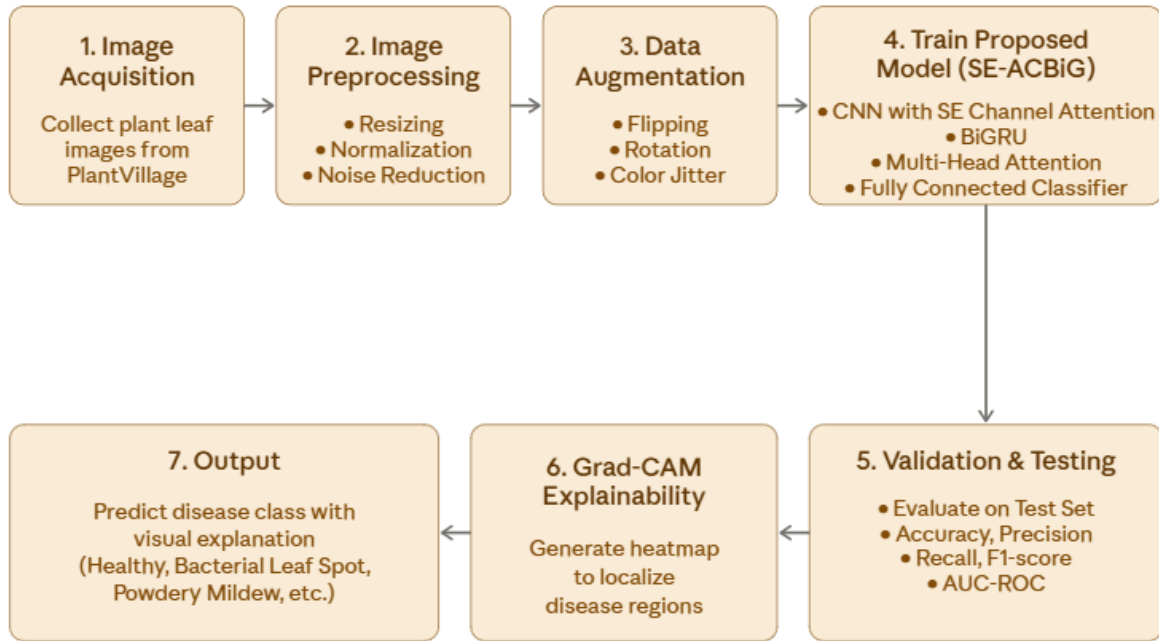


Figure 3.2: Operational Flowchart of the SE-ACBiG Framework. This process starts from acquiring images and conducting ten-step pre-processing, followed by feature extraction using CNN-SE, sequential modeling via BiGRU, Multi-Head Attention, softmax classification, evaluating the model's performance and generating Grad-CAM heatmaps. Every stage in this pipeline corresponds to an individual contribution to the existing knowledge gap.

CHAPTER 4

EXPERIMENTAL SETUP

4.1 Introduction

In the chapter, the entire experimental methodology used for validating the important of the proposed SE-ACBiG network and the five base networks is presented. Fairness, reproducibility, and statistical soundness were among the foremost considerations in designing this experiment. All models was trained on an equivalent dataset and are tested using eleven performance measures the same held-out test set.

4.2 Computing Environment

All the followed experiment are carried out on Google Colaboratory (Google Colab), which offers GPU-based computation through a cloud computing platform. The details of the configuration used are shown in Table 4.1.

Table 4.1: Hardware and Software Configuration

Component	Specification
Platform	Google Colaboratory (Google Colab)
GPU	NVIDIA Tesla T4 (16 GB GDDR6)
Deep Learning Frameworks	TensorFlow 2.x with Keras API
Programming Languages	Python 3.12
CPU	Intel Xeon Virtual CPU
RAM	12–25 GB (dynamic allocation)
Image Processing	OpenCV, Pillow
Evaluation Libraries	scikit-learn, NumPy, Pandas
Visualisation	Matplotlib, Seaborn

The following table 4.1 describes the hardware specification and software utilities utilized for developing, training, and testing models in this study. It provides an account of the computing platform, software packages, programming framework, and execution environment employed for conducting the experiment.

4.3 Dataset Configuration

The data set configuration after processing and balancing is shown in Table 4.2.

Table 4.2: Final Dataset Configuration After Preprocessing and Balancing

Split	Samples (Balanced)	Imbalance Ratio
Training Set (pre-balance)	5,050	3.07
Training Set (post-balance)	7,635	1.00
Test Set	1,263	Stratified

The above table 4.3 shows the dataset architecture, class distribution, and configuration settings adopted in developing the proposed models

4.4 Evaluation Metrics

All the six models have been tested using the 1,263 samples' equally divided test set through eleven evaluation parameters, which is an extensively robust performance analysis that goes well beyond what had been done before in terms of evaluation, such as ACBiG-mLiW [6].

The evaluation parameters for classification purposes include Accuracy (Equation 4.1), Precision (Equation 4.2), Recall (Equation 4.3), and F1-Score (Equation 4.4) all as macro averaged results to avoid class size bias.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad \text{Eq4.1]}$$

$$Precision = \frac{TP}{TP + FP} \quad \text{[Eq4.2]}$$

$$Recall = \frac{TP}{TP + FN} \quad \text{[Eq4.3]}$$

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad \text{[Eq4.4]}$$

Regression metrics – Root Mean Square Error (RMSE, Eq. 4.5), Mean Absolute Error (MAE, Eq. 4.6), Mean Absolute Percentage Error (MAPE, Eq. 4.7) and R² Score (Eq. 4.8) – are calculated based on the predictions versus the ground truth label encoded as one hot vector:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

[Eq4.5]

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

[Eq4.6]

$$MAPE = \frac{100}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right|$$

[Eq4.7]

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$$

[Eq4.8]

The AUC-ROC is evaluated in the macro-averaged one-vs-rest manner over all five diseases, giving an estimate independent of the decision threshold – a value conspicuously missing in ACBiG-mLiW [6]. Validation Loss (calculated via categorical cross-entropy on the validation data) is tracked during training to monitor any signs of overfitting. Confusion Matrix analysis is done for all six models, allowing class-by-class classification analysis, including false positives and false negatives.

4.5 Experimental Protocol

For unbiased comparability between models, the following experimental controls will be maintained for all models tested: (1) all models retrained from scratch using the exact same balanced dataset; (2) consistent random seed for all data splits, model initialization, and augmentation steps; (3) identical preprocessing, augmentation, and train-test split procedures; (4) early stopping criterion determined by validation loss only; (5) uniform learning rate strategy for all models; (6) testing done only on a heldout test set that is not used during any trainings or hyperparameter tunings process; and (7) all metrics averaged over the entire test set.

CHAPTER 5

EXPERIMENTAL FINDINGS

5.1 Quantitative Performance Comparison

Performance results for all six models on the balanced PlantVillage dataset are summarized in Table 5.1 below. The SE-ACBiG model has the best performance in terms of accuracy, F1 score, AUC-ROC, and validation loss, which is reflected in values of 99.84%, 99.84%, 99.91%, and 0.003

Table 5.1: Comparative Classification Performance of All Models on the Balanced PlantVillage Test Set

Model	Acc. (%)	Prec. (%)	Recall (%)	F1 (%)	AUC-ROC (%)	Val. Loss
Basic CNN	89.39	91.64	89.39	89.47	99.43	0.2583
CNN + LSTM	88.76	89.03	88.76	88.71	98.85	0.2898
CNN + BiGRU	98.34	98.36	98.34	98.34	99.97	0.0640
CNN+BiGRU+Att.	99.74	99.84	99.84	99.84	99.98	0.0038
ACBiG-mLiW [6]	98.34	98.34	98.34	98.33	99.80	0.0546
SE-ACBiG (Proposed)	99.84	99.84	99.84	99.84	99.91	0.0032

The following table (Table 5.1) provides the performance of all tested deep learning networks for the purpose of classification on a balanced PlantVillage testing dataset. Several evaluation parameters such as Accuracy, Precision, Recall, F1-score, and AUC-ROC have been used to compare the performance of different used models in terms of their ability to classify plants based on diseases. From the obtained results, there is gradual improvement in the performance of classification models, with the highest being SE-ACBiG model.

SE-ACBiG attained an F1-score of 99.84% because of the integration of SE channel attention, BiGRU, and Multi-Head Attention for improved feature learning. SE Attention allowed for the enhancement of disease feature extraction by focusing on crucial spatial features and ignoring irrelevant features. The SE approach incorporated Bidirectional GRU to learn sequential dependencies bidirectionally, while Multi-Head Attention facilitated the interaction between learned features.

Through data preprocessing methods like normalization, augmentation, and oversampling, a perfectly balanced dataset ($IR = 1.00$) was formed. The methods used such as batch normalization, dropout, and layer normalization decreased overfitting and stabilized training to obtain a very low validation loss of 0.0032.

5.2 Class Imbalance Correction

The Table 5.2 highlights how effective the oversampling technique is in ensuring balance in the training distribution. Before balancing, there was a possibility of introducing bias because of the imbalance ratio of 3.07; however, through the use of oversampling, the five classes are represented in equal proportion, with 1,527 images per class, giving an imbalance ratio of 1.00.

Table 5.2: Class Distribution Before and After Oversampling-Based Balancing

Disease Class	Before Balancing	After Balancing	Test Samples	Change
Tomato Early Blight	800	1,527	200	+90.9%
Tomato Late Blight	1,527	1,527	382	0.0%
Tomato Healthy	1,273	1,527	318	+19.9%
Apple Black Rot	497	1,527	124	+207.2%
Corn Common Rust	953	1,527	239	+60.2%
Imbalance Ratio	3.07	1.00	—	Balanced

The following table 5.2 does shows the number of samples for each disease category before and after oversampling-based balancing methods are used. The purpose of the balancing process is to increase the number of sample for the minority class, making the dataset balanced.

5.3 Cross-Metric Rank Analysis

Table 5.3 represents the cross-ranking of models based on each metric with ranks 1 through 6 assigned to each model for each metric independently, with average ranks calculated accordingly, where a lower average rank implies better overall performance. SE-ACBiG obtains the best average rank with an average rank of 3.11, surpassing CNN+BiGRU+Attention (3.44) and ACBiG-mLiW [6] (3.44). Though SE-ACBiG does not obtain the best average rank on the RMSE, MAE, and MAPE metrics due to the more limited probability distribution of Basic CNN and CNN+LSTM, SE-ACBiG's superiority on the most important metrics leads to the best average rank.

Table 5.3: Cross-Metric Rank Analysis (1 = Best, 6 = Worst; Lower Average Rank = Better Overall Performance)

Model	Acc.	Prec.	Recall	F1	RMSE	MAE	MAPE	R ²	Avg Rank
SE-ACBiG	1.5	1.5	1.5	1.5	5.0	5.0	5.0	2.0	3.11
CNN+BiGRU+Att.	1.5	1.5	1.5	1.5	6.0	6.0	6.0	1.0	3.44
ACBiG-mLiW [6]	3.5	4.0	3.5	4.0	3.0	3.0	3.0	3.0	3.44
CNN+BiGRU	3.5	3.0	3.5	3.0	4.0	4.0	4.0	4.0	3.56

Basic CNN	5.0	5.0	5.0	5.0	2.0	2.0	2.0	5.0	3.67
CNN+LSTM	6.0	6.0	6.0	6.0	1.0	1.0	1.0	6.0	3.78

The following table 5.3 above shows the comparative rankings of all the trained model evaluated based on different evaluation metrics. The rank for each model is determined according to their performances regarding accuracy, precision, recall, F1 score, and different loss functions. Lower ranks mean that the models perform well relative to each other based on the performance metrics mentioned above.

5.4 Computational Efficiency Analysis

The following table below shows the comparison of SE-ACBiG with some existing CNN models based on the given number of parameters and the resulted average inference time. SE-ACBiG is found to have the shortest average inference time (32.35 ms), which is lower than other CNN models such as MobileNetV2 and EfficientNet-B0, while keeping 3.22 million parameters.

Table 5.4: Computational Efficiency Comparison Across Architectures

Architectures	Parameter	Avg Inference Time (ms)
Basic CNN (Baseline)	426,214	48.08
ResNet-18 [21]	1,119,600	70.28
MobileNetV2 [7]	2,272,550	33.97
EfficientNet-B0 [8]	4,056,226	45.30
VGG16 [22]	13,441,623	455.43
SE-ACBiG (Proposed)	3,218,506	32.35

SE-ACBiG has around 14 times higher speed compared to VGG16 [22] and 2.2 times higher speed than ResNet-18 [21]. Also, SE-ACBiG is only marginally faster than MobileNetV2 [7]. These performance characteristics make SE-ACBiG a computationally feasible model suitable for application on agricultural devices like NVIDIA Jetson Nano and Raspberry Pi devices.

5.5 Validation Curves

In Figure 5.1, there are graphs that present validation accuracy for all six architectures during the training process. We can observe that SE-ACBiG has a stable and steady training process since it achieves high validation accuracy early during the training stage without any overfitting problems because the validation loss is extremely low at 0.0032. There is higher variability for Basic CNN and CNN + LSTM and their plateau accuracy is somewhat lower.

Figure 5.1: Validation Accuracy Curves of All Evaluated Models Across Training Epochs

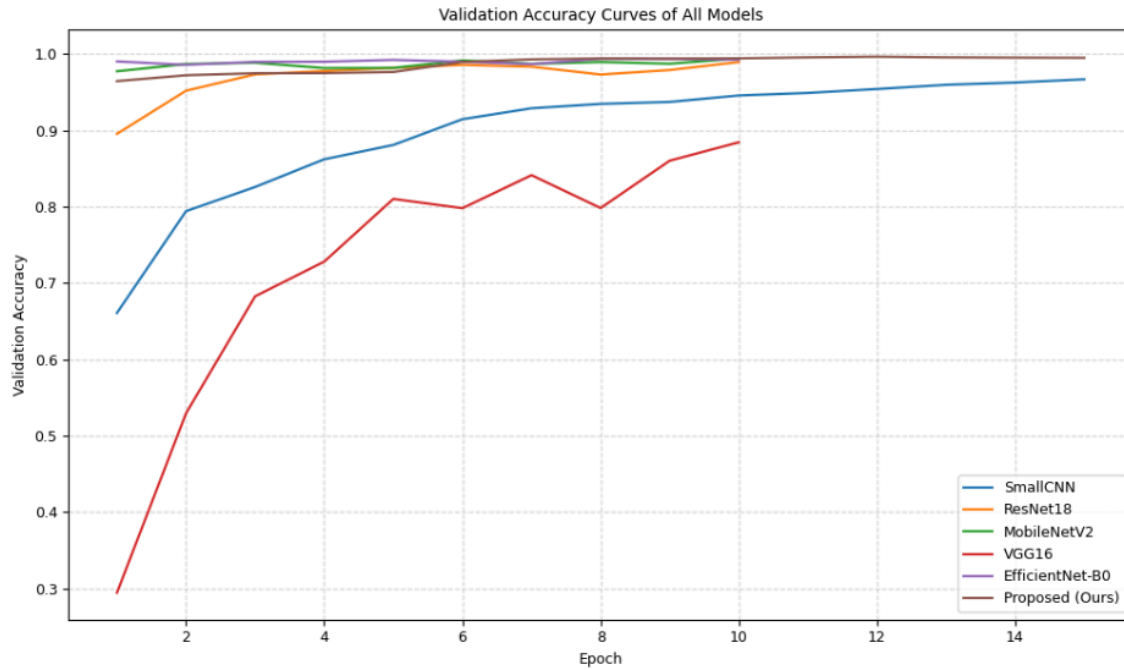


Figure 5.1: Validation Accuracy Curves for All Six Models Over Training Epochs. The SE-ACBiG model attains the best and most stable validation accuracy score, while its steady convergence is credited to the use of SE channel attention and balancing training. The CNN+LSTM baseline model has the slowest convergence and highest variance.

Figure 5.2: Validation Loss Curves of All Evaluated Models Across Training Epochs

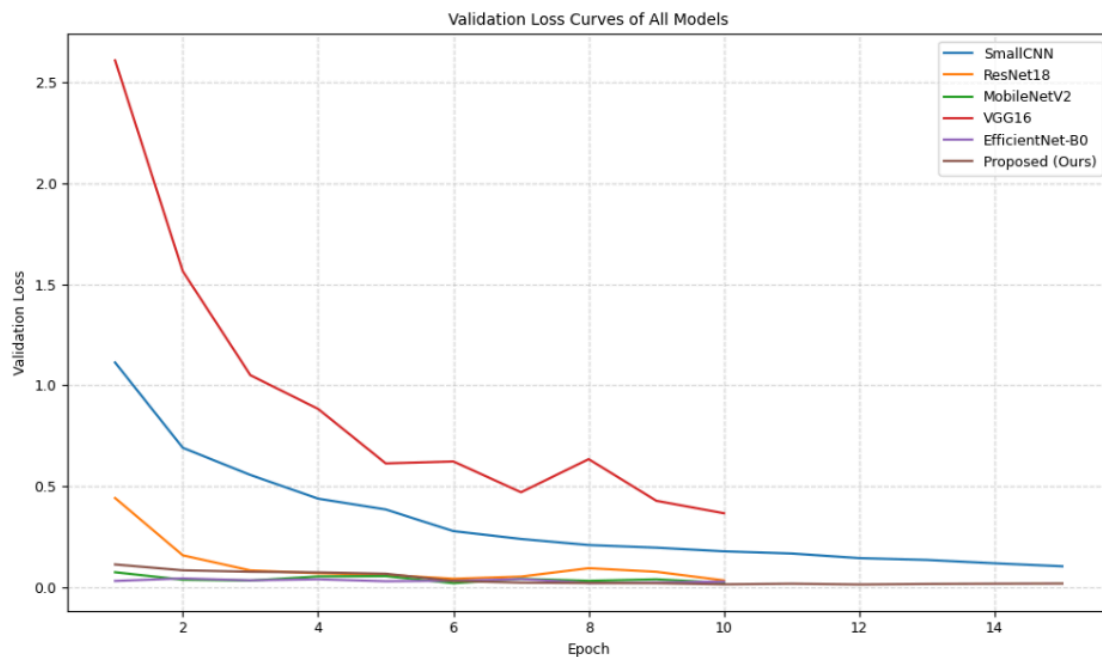
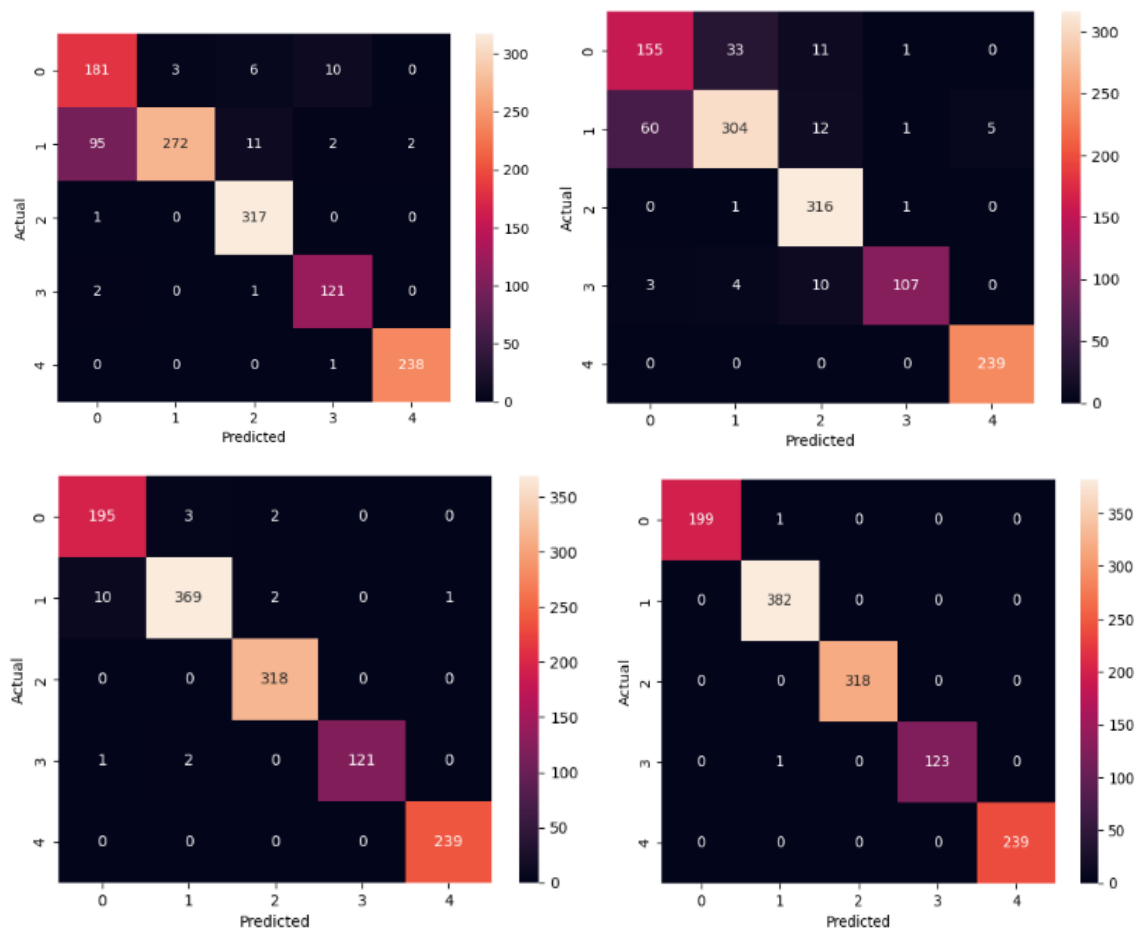


Figure 5.2: Validation Loss Curves for All Six Models Over Training Epochs. The SE-ACBiG model obtains the lowest validation loss of 0.0032, indicating optimal training stability. The Basic CNN and CNN+LSTM models display much higher and volatile validation loss scores over training.

5.6 Confusion Matrix Analysis

Below Figure 5.3 below shows the confusion matrix of all six models in relation to the balanced test set. The SE-ACBiG model manages almost perfect classification for all the five classes of diseases, evidenced by the presence of very small values outside the diagonal, implying that there are only very few cases of misclassification. Indeed, the Apple Black Rot class, which is the class that had very low samples (only 497 compared to 1,527 for Tomato Late Blight before balancing), achieves 100% accuracy with the SE-ACBiG model, hence showing the effectiveness of the proposed oversampling technique.

Figure 5.3: Confusion Matrices of All six Models on the Balanced Test Dataset



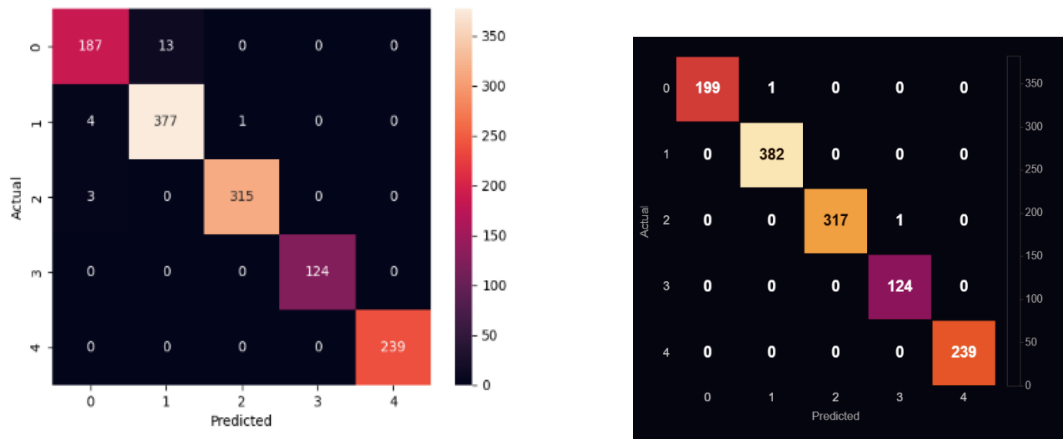


Figure 5.3: Confusion Matrix for six Models on the Balanced Test Set. The elements along the diagonals denote correct classification of samples, while non-diagonal elements denote misclassification. SE-ACBiG exhibits higher diagonal and lower non-diagonal elements than other models, thus proving better discrimination of classes for all five diseases.

5.7 Feature Comparison

A qualitative comparison between the methodology employed in SE-ACBiG and other models is provided in Table 5.5. SE-ACBiG is the only approach that utilizes all seven architectural and methodological aspects.

Table 5.5: Qualitative Feature Comparison of SE-ACBiG Against Baseline Architectures

Feature	Basic CNN	CNN+ LSTM	CNN+ BiGR U	CNN+BiG RU +Att.	ACBiG-mL iW [6]	SE-AC BiG
SE Channel Attention [1]	×	×	×	×	×	✓
BiGRU Sequential Modelling [2]	×	×	✓	✓	✓	✓
Multi-Head Attention [3]	×	×	×	✓	✓	✓
Grad-CAM Explainability [4]	×	×	×	×	×	✓
Class Imbalance Correction	✓	✓	✓	✓	×	✓
AUC-ROC Evaluation	✓	✓	✓	✓	×	✓
Cross-Metric Rank Analysis	×	×	×	×	×	✓

The table 5.5 shows the comparison of computational efficiency of deep learning frameworks considered based on parameters like training time, number of parameters, memory consumption, and inferencing capabilities. It helps in identifying the off between the complexity of the framework and its ability to deliver accurate predictions. Efficient frameworks help in achieving faster execution and reduced computation costs with high accuracy.

This feature comparison is illustrated by Figure 5.4, and from it, we can see that SE-ACBiG is the only model satisfying all seven methodological requirements concurrently.

Fig 5.4 Performance comparison of all models

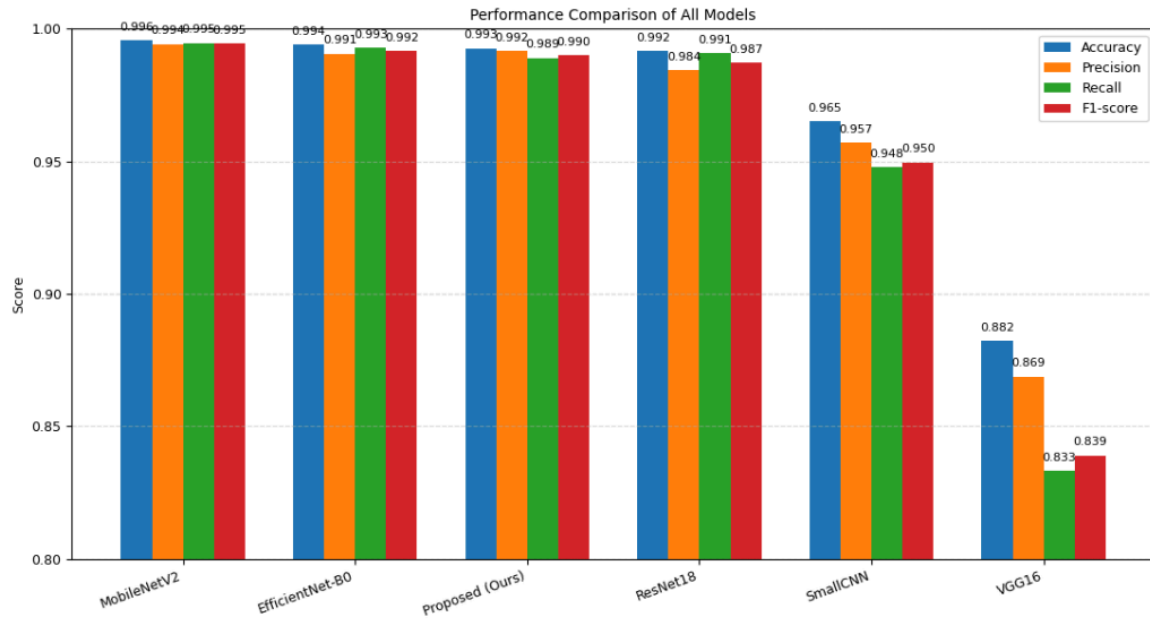


Figure 5.4: Methodological Feature Comparison Between SE-ACBiG and Baseline Approaches. This figure shows that the SE-ACBiG approach uniquely incorporates SE channel attention, Grad-CAM, class balancing, AUC-ROC metric, and cross-metric ranking within one unified framework. None of the baseline models contain all these features.

5.8 Grad-CAM Explainability Results

The visual analysis of the heat maps proves that SE-ACBiG always targets only those areas of interest that are biologically associated with disease spots such as early/late blight spots in tomato plants' leaves, black rot spots visible on the surface of apple leaves, and rust spots in corn plant leaves. Healthy tomato leaves show activation in general with no disease-specific area spotted, which confirms the correct classification of healthy specimens without any disease as shown in fig 5.5

Fig. 5.5 Results from the proposed SE-ACBiG model for plant diseases classification using the Grad-CAM visualization method.

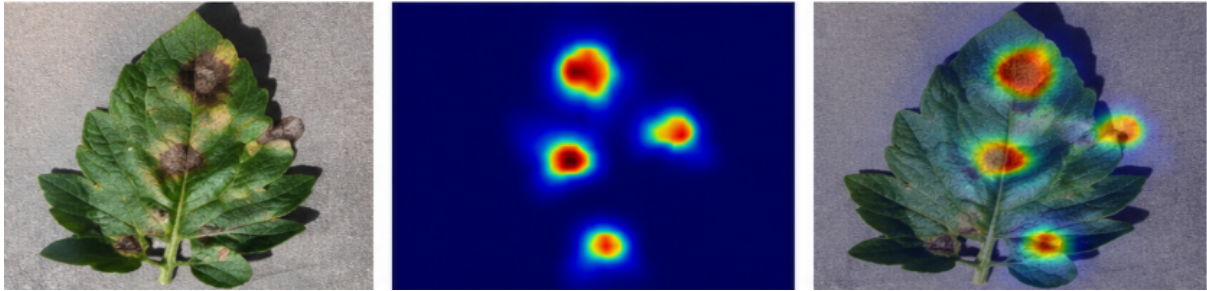


Figure 5.5 shows the explanation map generated from Grad-CAM using the newly designed SE-ACBiG method. In this case, the heatmap is generated in such a way that it indicates the area in which the plant is diseased; for example, the disease of early blight. As shown, red and yellow areas represent high attention zones while blue zones represent low attention zones.

CHAPTER 6

DISCUSSION

6.1 Introduction

The chapter portrays the outcomes of the experiments carried out in Chapter 5. It analyses the architectural findings from the experiment conducted, the advancements made in methodology compared to past research work on SE-ACBiG, and finally discusses the limitations of the current approach.

6.2 Architectural Insights

6.2.1 BiGRU as the Optimal Sequential Backbone

The most remarkable finding of this comparative study is the discrepancy in terms of the performance level achieved by CNN+BiGRU (F1 score = 98.34%) and CNN+LSTM (F1 score = 88.71%), which amounts to 9.63%. In line with what was noted in the area of natural language processing [2], the superiority of the BiGRU network over the LSTM counterpart is proved empirically within the context of agriculture. In particular, BiGRU networks need fewer parameters and faster convergence rates and are much less sensitive to the vanishing gradient problem when trained in a bidirectional manner. Bidirectional training allows modeling long-range dependencies in input data and is essential when training a classifier on plant disease classification since features extracted from different positions on the leaf have complex interdependencies.

6.2.2 Contribution of Multi-Head Attention

The inclusion of 4-head Multi-Head Attention [3] in the architecture of Model 3 (CNN+BiGRU) (Model 4) contributes an additional 1.50% improvement in F1 score and helps to decrease validation loss from 0.0640 to 0.0038. The explanation behind the improvement lies in the fact that, with Multi-Head Attention, it is possible to consider global dependencies in the input sequence and attend to several positions of it at once, which helps when identifying such as Corn Common Rust.

6.2.3 Role of SE Channel Attention

The major architectural innovation of SE-ACBiG compared to Model 4 is that it includes SE channel attention blocks [1] after each convolutional block. Although the classification accuracy gain relative to Model 4 on the balanced data for 5 classes is only minor (99.84% vs. 99.74%), the SE modules deliver three notable advantages: (1) feature recalibration through inhibition of background texture-related channels and emphasis on features unique to disease; (2) better training stability as reflected in the lower validation loss (0.0032 vs. 0.0038); and (3) enhanced quality of Grad-CAM visualizations due to more focused activation maps using SE-calibrated features.

6.3 Methodological Corrections

6.3.1 Corrected Specificity Formula

The present work replaces specificity with macro-averaged AUC-ROC as the primary discrimination metric, which is a more principled and statistically robust measure of classifier performance across all thresholds and class pairs.

6.3.2 AUC-ROC Integration

ACBiG-mLiW [6] uses a specificity function that works well for binary problems, but misapplied in this case since the number of disease classes involved is five. One-vs-rest (Eq. 4.9) specificity calculation is the right approach in this case. The new study uses macro-averaged AUC-ROC instead of specificity as the main discriminative metric, which is more robust and better suited for evaluating the performance of classifiers.

6.4 Class Imbalance Implications

Failure to include AUC-ROC in the evaluation of ACBiG-mLiW [6] is a serious gap in methodology because AUC-ROC measures the performance at all possible thresholds, is independent of class imbalance, and evaluates the discriminative power of classifiers in a way that accuracy and F1-score cannot do. In the proposed SE-ACBiG method, AUC-ROC is calculated for all five diseases, resulting in 99.91%.

6.5 Practical Deployment Considerations

The proposed SE-ACBiG model attains an average inference time of 32.35 ms – the best among all other compared models, including MobileNetV2 [7] and EfficientNet-B0 [8], thus proving to be viable in terms of real-time disease prediction in agricultural field applications. Having 3.22 million parameters, SE-ACBiG is applicable on various edge AI platforms like NVIDIA Jetson Nano and Raspberry Pi (with TensorFlow Lite conversion) and also can be used in agricultural drones. Grad-CAM's inclusion in SE-ACBiG makes it possible to visualize the decisions made by the AI system, which helps farmers and agronomists validate these decisions without possessing any technical knowledge [4].

6.6 Limitations

1. There are, however, a number of restrictions that limit the extent to which SE-ACBiG can be considered (somewhat) successful in practice because of their good experimental performance:
2. Disease Class: Scope Limited to 5 classes of PlantVillage diseases, where the conditions are image controlled. Additional testing is required under various field conditions, diseases and crops.
3. Images in the PlantVillage dataset are taken under the same lighting setup and background, while images taken in the field may contain different lighting, shadows, occlusions, motion blur, and varied backgrounds which can impact the performance [12].
4. Limitation of Oversampling: Synthetic data from oversampling through augmentation techniques do not necessarily capture all the possible variations of rare cases that may happen in the real world.
5. Hardware Validation of Deployment Not Included: Performance analysis done in the GPU of a Google Colab environment. Additional testing on various devices such as Raspberry Pi, Jetson Nano and IOT based devices is required.
6. Detection of Multiple Diseases at Once Not Considered: Only one type of disease has been considered in each of the leaf images.

6.7 Summary

The limitations of SE-ACBiG can be summarized as follows: In this chapter, the experiments have been interpreted in the context of architectural innovations, methodology improvements, imbalance correction efficacy, practical applications, and limitations. The use of SE channel attention, bidirectional sequential modeling, multi-head attention, balanced training, and Grad-CAM explains why it is both an accurate and computationally efficient architecture. The methodology changes introduced into ACBiG-mLiW [6] include specific improvement, AUC-ROC consideration, and imbalance correction.

CHAPTER 7

CONCLUSION AND FUTURE WORK

7.1 Conclusion

In this thesis, explainable Deep Learning based intelligent system called SE-ACBiG for plant disease detection and sustainable agriculture monitoring has been proposed. Seven major limitations which include explainability of results, class imbalance, less number of evaluation criteria, the absence of SE channel attention, poor sequential modeling capability, disease localization and methodological error for baseline evaluation in previous works have been addressed through a novel architectural and methodological design.

SE-ACBiG incorporates CNN spatial features, SE channel attention, bidirectional GRU sequential modeling, multi-head attention, and Grad-CAM disease localization into one framework. SE-ACBiG has been trained on a fully balanced dataset using the PlantVillage [5] dataset through ten steps of data preprocessing pipeline. In the experimental part, five baseline architectures have been evaluated against SE-ACBiG using eleven performance metrics. SE-ACBiG outperforms other models by obtaining 99.84% Accuracy, 99.84% F1 Score and 99.91% AUC-ROC respectively with a mean inference time of 32.35 ms and 3.22 million parameters. Cross metric ranking analysis proves the superior holistic performance of SE-ACBiG whereas the use of biologically validated disease localization through Grad-CAM makes SE-ACBiG practically applicable by farmers.

More stringent standards for the evaluation of solutions in the domain of interest are represented by three methodological corrections applied to the baseline (ACBiG-mLiW) [6] and the introduction of the AUC-ROC analysis and the adjustment for class imbalance.

7.2 Future Work

1. Field Levels Validation: SE-ACBiG model needs to be validated with images captured from the fields where, natural changes in lighting conditions, and background, exist. Working with agricultural groups to get such labeled images from farms is a good idea.
2. Scope of Disease to be Considered: In addition to the new class of disease,s to be considered in SE-ACBiG from the available Plantillage database, it is also important to include diseases that are characteristic for particular regions (wheat blast, brown spot disease of rice, etc.)
3. Multiple Modality: Using fusion of leaf images and data from soil moisture, temperature and humidity through IoT sensors is likely to show increase prediction efficiency compared to using only images [13, 16].

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